

CCNP Lab: Layer 2 Attacks and Mitigations



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Purpose:

This lab aims to set up three different layer 2 attacks and then build a network environment that fortified against them: DHCP starvation, ARP spoofing, and MAC flooding. To mitigate these threats, we implemented the following security measures: DHCP Snooping to protect against DHCP starvation, Dynamic ARP Inspection (DAI) to guard against ARP spoofing, and Port Security to counter MAC flooding.

We actively launched these three types of attacks from a Linux server against a network switch. offense and defense type learning allows us to deepen our understanding of how Layer 2 vulnerabilities can be exploited and how to best defend against them. Ultimately, the exercise is designed to enhance our ability to secure enterprise networks and apply Layer 2 protection strategies in real-world networking environments.

Background Information on lab concepts:

DHCP Starvation Attack: This is a denial-of-service technique in which an attacker rapidly generates DHCP requests using spoofed MAC addresses, exhausting the DHCP server's available IP address pool and preventing legitimate devices from obtaining network access.

ARP Spoofing: ARP spoofing occurs when an attacker sends falsified ARP messages over a network. This tricks devices into associating the attacker's MAC address with the IP address of another device (e.g., a gateway), enabling interception or disruption of traffic.

MAC Flooding: This attack targets a switch's MAC address table by overwhelming it with fake MAC addresses. Once the table overflows, the switch enters a fail-open mode, forwarding all frames to every port and compromising network segmentation.

Dynamic ARP Inspection (DAI): DAI prevents ARP spoofing by inspecting ARP packets on untrusted ports and validating their consistency with trusted IP-to-MAC bindings derived from DHCP Snooping or static configuration.

Port Security: This switch-level feature limits the number of MAC addresses that can be learned dynamically on a port, allowing administrators to control which devices are permitted to connect and respond to violations by shutting down the port or restricting access.

DHCP Snooping: This feature filters unauthorized DHCP messages by allowing DHCP communication only on trusted ports. It builds a database of legitimate bindings and drops traffic from untrusted sources, effectively preventing rogue DHCP servers and DHCP exhaustion.

Lab Summary:

In this lab, we built a small Layer 2 network — a PC, a Cisco switch, a Cisco router acting as the DHCP server and gateway, and a Kali Linux server used as the attacker — and used it to both launch and defend against three Layer 2 attacks. From the Linux server we performed a MAC flooding attack with macof, a DHCP starvation attack that exhausted the router's DHCP pool, and an ARP spoofing attack that poisoned the gateway's ARP table. We confirmed each attack succeeded by inspecting the switch's MAC address table and the router's ARP table for the tell-tale evidence (an overflowed MAC table, "Incomplete" ARP entries from pool exhaustion, and a duplicated MAC address bound to two IPs, respectively). We then configured the matching mitigation on the switch for each attack (Port Security against MAC flooding, DHCP Snooping against DHCP starvation, and Dynamic ARP Inspection against ARP spoofing) and re-ran each attack to confirm it was now blocked.

Network Diagram and Topology:

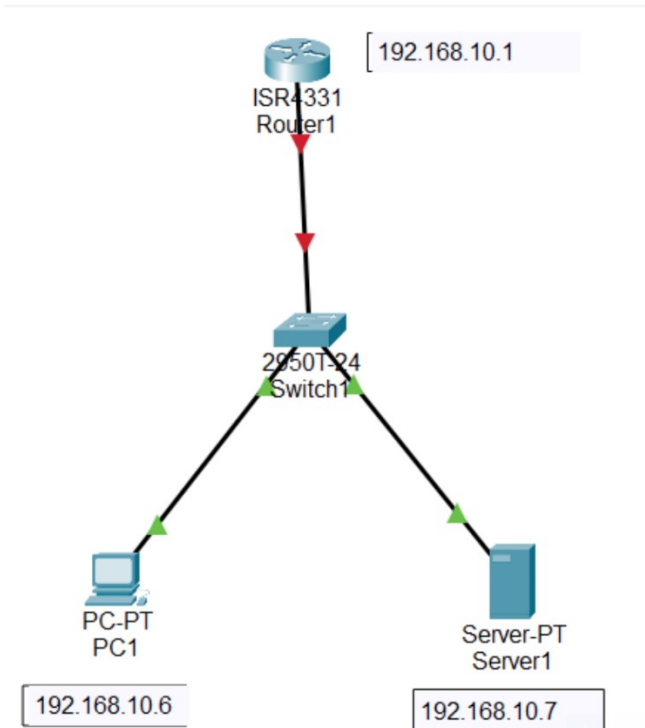
PC (Connects to Switch through F0/1) (192.168.10.6)

Switch (Connects to Router through F0/2)

Router (Connects to Switch through Interface G0/0/0) (192.168.10.1)

Linux Server (Connects to Switch through F0/3) (192.168.10.7)

PC to switch, switch to router, Linux server to switch. Linux server used to run the attacks on the switch.



Lab Commands:

show ip route – Displays the IPv4 routing table

ip routing – A command to enable the sending of data packets for IPv4 across an IP network

tracert [destination ip address] - Figuring the data hops from the source ip address to the destination ip address.

show ip protocol – displays information about routing protocols on the device

ip arp inspection vlan [vlan_number] -used to enable Dynamic ARP Inspection (DAI) on specific VLANs within a network

ip arp inspection trust – used to configure an interface trusted for Dynamic ARP Inspection (DAI) on a Switch.

ip dhcp snooping – to enable DHCP snooping globally on a Cisco Switch

ip dhcp snooping vlan – enables DHCP snooping on specific VLANs on a switch

ip dhcp snooping trust – designates a switch port as a trusted interface for DHCP traffic

ip dhcp snooping limit rate [number] - sets the number of dhcp requests that can be revived in a second.

sudo macof -i <interface> - enables mac flooding on interface

switchport mode access – configures a switchport to operate in access mode

Switchport port-security - a security feature on network switches that restricts network access to authorized devices by limiting the number of MAC addresses allowed on a port.

Router configuration:

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ip dhcp excluded-address 192.168.10.1
ip dhcp pool LAYER2ATTACK
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
login on-success log
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.10.1 255.255.255.0
negotiation auto
interface GigabitEthernet0/0/1
no ip address
negotiation auto
interface GigabitEthernet0/2/0
```

```
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Switch configuration:

```
version 12.2
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption

hostname Switch

boot-start-marker
boot-end-marker

no aaa new-model
system mtu routing 1500
ip arp gratuitous none

ip dhcp snooping

spanning-tree mode pvst
spanning-tree portfast default
spanning-tree extend system-id

vlan internal allocation policy ascending

interface Vlan1
no ip address
shutdown
```

```

ip http server
ip http secure-server
logging esm config
line con 0
line vty 5 15
end

```

PC configuration:

```

Description . . . . . : Intel(R) Ethernet Connection
(17) I219-LM
Physical Address. . . . . : 04-D9-C8-BA-24-6E
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv4 Address. . . . . : 192.168.10.6(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Monday, June 9, 2025 10:02:17
AM
Lease Expires . . . . . : Tuesday, June 10, 2025
11:03:25 AM
Default Gateway . . . . . : 192.168.10.1
DHCP Server . . . . . : 192.168.10.1
DNS Servers . . . . . : 8.8.8.8
NetBIOS over Tcpip. . . . . : Enabled

```

Show IP route/Traceroute for IPv4:

Router:

```
Router#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2

ia - IS-IS inter area, * - candidate default, U - per-user
static route

o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP

a - application route
+ - replicated route, % - next hop override, p - overrides
from PfR

Gateway of last resort is not set

```
          192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C          192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L          192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
```

Show IP protocol:

```
Router#show ip protocol
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)
```

MAC Flooding Proof:

Mac Address Table

Vlan	Mac Address	Type	Ports
----	-----	-----	-----
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0100.0ccc.ccce	STATIC	CPU
All	0180.c200.0000	STATIC	CPU
All	0180.c200.0001	STATIC	CPU
All	0180.c200.0002	STATIC	CPU
All	0180.c200.0003	STATIC	CPU

All	0180.c200.0004	STATIC	CPU
All	0180.c200.0005	STATIC	CPU
All	0180.c200.0006	STATIC	CPU
All	0180.c200.0007	STATIC	CPU
All	0180.c200.0008	STATIC	CPU
All	0180.c200.0009	STATIC	CPU
All	0180.c200.000a	STATIC	CPU
All	0180.c200.000b	STATIC	CPU
All	0180.c200.000c	STATIC	CPU
All	0180.c200.000d	STATIC	CPU
All	0180.c200.000e	STATIC	CPU
All	0180.c200.000f	STATIC	CPU
All	0180.c200.0010	STATIC	CPU

Total Mac Addresses for this criterion: 20

DHCP Starvation Proof:

```
Router# show ip arp
```

Protocol	Address	Age (min)	Hardware Addr	Type
Interface				
Internet	192.168.10.1	-	cc7f.76d1.adc0	ARPA
GigabitEthernet0/0/0				
Internet	192.168.10.6	0	04d9.c8ba.246e	ARPA
GigabitEthernet0/0/0				
Internet	192.168.10.7	2	482a.e38e.e9c8	ARPA
GigabitEthernet0/0/0				
Internet	192.168.10.230	0	Incomplete	ARPA
Internet	192.168.10.231	0	Incomplete	ARPA
Internet	192.168.10.232	0	Incomplete	ARPA
Internet	192.168.10.233	0	Incomplete	ARPA
Internet	192.168.10.234	0	Incomplete	ARPA
Internet	192.168.10.235	0	Incomplete	ARPA
Internet	192.168.10.236	0	Incomplete	ARPA
Internet	192.168.10.237	0	Incomplete	ARPA
Internet	192.168.10.238	0	Incomplete	ARPA
Internet	192.168.10.239	0	Incomplete	ARPA
Internet	192.168.10.240	0	Incomplete	ARPA
Internet	192.168.10.241	0	Incomplete	ARPA
Internet	192.168.10.242	0	Incomplete	ARPA

Internet	192.168.10.243	0	Incomplete	ARPA
Internet	192.168.10.244	0	Incomplete	ARPA

ARP Spoofing Proof:

```
Router# show ip arp
Protocol Address          Age (min)  Hardware Addr  Type
Interface
Internet  192.168.10.1            -    cc7f.76d1.adc0  ARPA
GigabitEthernet0/0/0
Internet  192.168.10.6            0    482a.e38e.e9c8  ARPA
GigabitEthernet0/0/0
Internet  192.168.10.7            5    482a.e38e.e9c8  ARPA
GigabitEthernet0/0/0
```

Traceroute:

Router to PC

```
Router#ping 192.168.10.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
Router#
```

PC to Router

```
C:\Users\Andy> ping 192.168.10.1
```

Pinging 192.168.10.1 with 32 bytes of data:

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

Router to Server

Router# ping 192.168.10.7

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.10.7, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

Problems:

What was particularly challenging (half page):

The most difficult aspect of the lab involved executing the DHCP starvation and ARP spoofing attacks. Unlike MAC flooding, which required just a single command (macof), the other attacks demanded more nuanced setup and validation. Verifying that ARP spoofing was successful was particularly tricky—it required analyzing the ARP table to confirm that both the PC and Linux server were bound to the same MAC address. Similarly, confirming a successful DHCP starvation attack involved identifying "incomplete" entries in the ARP table, indicating that the DHCP pool was exhausted. However, even though MAC flooding was easy to start and defend, it was hard to show that it was actually working as intended. It was a struggle for the switch to be intercepting packets from other device trying to communicate with other devices.

Concepts we learned (half page):

Throughout the lab, we gained hands-on experience performing and identifying the three major Layer 2 attacks. We learned that MAC flooding manifests as an overflowed MAC table, DHCP starvation results in incomplete address bindings, and ARP spoofing causes duplicate MAC address associations for different IPs. Furthermore, we learned to configure defensive mechanisms such as DHCP Snooping, Port Security, and Dynamic ARP Inspection to detect and prevent these attacks in real time.

What caused issues (half page):

A major issue was the sequencing of attack demonstrations alongside their corresponding mitigations. While we could demonstrate each attack and its prevention independently, running them consecutively posed difficulties. Resetting the switch state and verifying that the security configurations remained effective added complexity. Another significant challenge was collecting solid evidence to verify that the attacks succeeded—especially in the case of ARP spoofing, where proof required recognizing anomalous MAC-to-IP mappings in the ARP table, as well as Mac flooding, which required the use of Wireshark to attempt to capture illegal packets (packets from other hosts communicating with another host, router, etc).

Conclusion:

In this lab, we successfully executed and mitigated three common Layer 2 attacks: DHCP starvation, ARP spoofing, and MAC flooding. Each attack targeted critical vulnerabilities at the data link layer, demonstrating how adversaries can disrupt normal network operations or compromise data integrity.

To protect against these threats, we used several security features: **DHCP Snooping** to prevent unauthorized IP address allocation, **Dynamic ARP Inspection (DAI)** to validate ARP packets and stop ARP-based impersonation, and **Port Security** to limit MAC address learning and prevent MAC flooding.

By implementing both offensive and defensive components, we gain a knowledge a comprehensive understanding of how Layer 2 attacks operate and how they can be detected and protected against using Cisco network configurations. This knowledge is essential for designing secure enterprise networks and strengthening infrastructure against internal and external threats.

Teacher Signoff Page of Lab Complete:

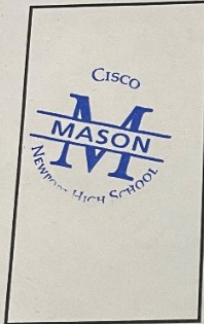
Layer 2 Attacks

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P3-4 CCNP

Mr. Mason

DHCP Starvation



ARP Spoofing



MAC Flooding

